

73-103 · CMU-Q

# Principles of Macroeconomics

Fall 2026

Andrés Castaño Zuluaga  
Carnegie Mellon University Qatar  
Business Administration

## Measuring Prices

---

# Course roadmap

|                  | Topic                    | Description                                                | Tools                                  | Use                              |
|------------------|--------------------------|------------------------------------------------------------|----------------------------------------|----------------------------------|
| Measurement      | <b>GDP</b>               | What the economy produced                                  | Circular flow, $Y \equiv C+I+G+NX$     | Read a release; Qatar energy GDP |
|                  | <b>Prices</b>            | Persistent rise in the price level                         | CPI / deflator, real vs nominal        | Contract, wage, peg              |
|                  | <b>Labor</b>             | Employment, unemployment, participation                    | Labor-force identities; short AI/labor | GCC hiring / participation       |
| Short-run demand | <b>Goods market</b>      | What determines demand?<br>Equilibrium and the multiplier. | ZZ, multiplier, Blanchard 3            | Fiscal impulse                   |
|                  | <b>Financial markets</b> | How $i$ is set                                             | Money market, Blanchard 4              | QCB / Fed; $i = \bar{i}$         |
|                  | <b>IS-LM</b>             | Goods + financial together                                 | IS, LM, policy mix, dashboards         | Shock → which curve, which way   |
| Medium run       | <b>Phillips</b>          | Inflation and unemployment together                        | Phillips curve, Blanchard 7–8          | Wage / inflation tradeoff        |
|                  | <b>Growth</b>            | Why levels differ                                          | Production function, growth accounting | Handout 4 + R                    |
| Open             | <b>Open economy</b>      | Trade, rates, and the peg                                  | NX, Blanchard 17–19                    | Shock → NX / peg (Qatar)         |

Midterm · Debate

# Class roadmap

## 1 Motivation

Why measuring prices matters

## 2 Causes of Inflation

Cost-push, demand-pull, expectations, monetary

## 3 Quantity Theory

$M \times V = P \times Y$  framework

## 4 Active Learning 1

Applying the Quantity Theory

## 5 Measuring Inflation

CPI and other indexes

## 6 CPI Construction

Step-by-step calculation

## 7 Active Learning 2

CPI calculation exercises

## 8 Problems with CPI

Biases and limitations

## 9 Key Takeaways

Summary and conclusions

Section 01

# Motivation

# Why Measuring Prices Matters

A price index is an input to decisions, not just a statistic.

| WHO           | THE DECISION                     | WHY THE PRICE INDEX MATTERS                                      |
|---------------|----------------------------------|------------------------------------------------------------------|
| Central banks | Set the policy interest rate     | The inflation target (~2%) is defined on a measured price index. |
| Governments   | Adjust benefits and tax brackets | Cost-of-living adjustments (COLA) track the official index.      |
| Businesses    | Set prices and wages             | Real profit margins depend on input vs. output prices.           |
| Workers       | Negotiate wages                  | Purchasing power is the real wage, not the nominal wage.         |
| Investors     | Allocate the portfolio           | What matters is the real return: nominal return minus inflation. |

**If the index is biased, every one of these decisions is calibrated with misguided information.**

Section 02

# Causes of Inflation

# Understanding the Causes of Inflation

## Cost-Push Inflation

Rising production costs push prices up.

*Example: Oil price shocks increase transportation and manufacturing costs.*

## Demand-Pull Inflation

Excess demand relative to supply.

*Example: Post-pandemic spending surge with supply constraints.*

## Expectation-Driven Inflation

Self-fulfilling price expectations.

*Example: Workers demand higher wages expecting future inflation.*

## Monetary Inflation

Excess money supply growth.

*Example: Central bank expanding money supply faster than economic growth.*

### WHAT IS MONEY SUPPLY?

- Total amount of money in an economy
- Includes cash, coins, and bank deposits
- Measured and controlled by central banks



### Key Concept

The **Quantity Theory of Money (QTM)** provides the mathematical framework for understanding monetary inflation.

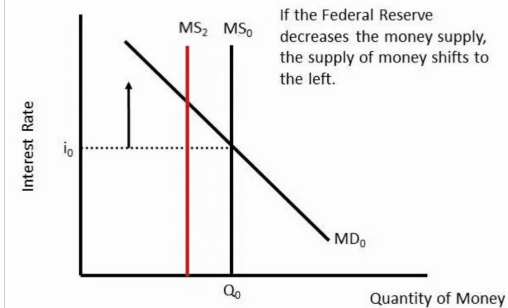
# The Quantity Theory of Money

## The Equation of Exchange

$$M \times V = P \times Y$$

| Var      | Definition        | Example                  |
|----------|-------------------|--------------------------|
| <b>M</b> | Money Supply      | Currency + Deposits      |
| <b>V</b> | Velocity of Money | Rate money changes hands |
| <b>P</b> | Price Level       | Avg price of goods       |
| <b>Y</b> | Real Output (GDP) | Quantity of goods        |

## Decreases in the Supply of Money



## Growth Rate Form

$$\% \Delta M + \% \Delta V = \% \Delta P + \% \Delta Y$$

## Solving for Inflation (% $\Delta P$ )

$$\text{Inflation} = \% \Delta M + \% \Delta V - \% \Delta Y$$

## Key Insight

If velocity (V) is stable, then inflation equals money supply growth minus real GDP growth. This links monetary policy directly to price levels.



Active Learning 01

# Applying the Quantity Theory of Money

# Active Learning 1

## Monetary Policy and Inflation

### Scenario

US central bank (Federal Reserve) is analyzing inflation pressures. Consider the following economic data:

|                                       |    |
|---------------------------------------|----|
| Money supply growth ( $\% \Delta M$ ) | 8% |
| Velocity of money ( $\% \Delta V$ )   | 0% |
| Real GDP growth ( $\% \Delta Y$ )     | 3% |

### Questions

1

According to the Quantity Theory of Money, what is the predicted inflation rate?

2

If the central bank wants to achieve a 2% inflation target, what should be the money supply growth rate?

3

What happens to inflation if real GDP growth slows to 1% while money supply growth stays at 8%?

Recall: **Inflation ( $\% \Delta P$ ) =  $\% \Delta M + \% \Delta V - \% \Delta Y$**

# Active Learning 1

## Solution

### Question 1: Predicted Inflation Rate

$$\text{Inflation} = 8\% + 0\% - 3\% =$$

5%

### Question 2: Money Supply for 2% Target

$$2\% = \% \Delta M - 3\% \rightarrow \% \Delta M = 2\% + 3\%$$

5%

### Question 3: Impact of Slower GDP Growth

$$\text{Inflation} = 8\% + 0\% - 1\% =$$

7%

*Note: Slower growth (1% vs 3%) with the same money supply leads to higher inflation.*

### Key Takeaway

**Slower economic growth with unchanged money supply growth leads to higher inflation. This explains why central banks often tighten monetary policy when growth slows unexpectedly.**

Section 03

# How Is Inflation Measured?

# Three Key Price Indexes

## 1 Consumer Price Index (CPI)

Measures the overall cost of goods and services bought by an average consumer.

**Primary use:** cost-of-living adjustments and inflation reporting.

## 2 Core CPI

The CPI excluding food and energy prices, which are volatile.

**Primary use:** identifying underlying inflation trends.

## 3 Producer Price Index (PPI)

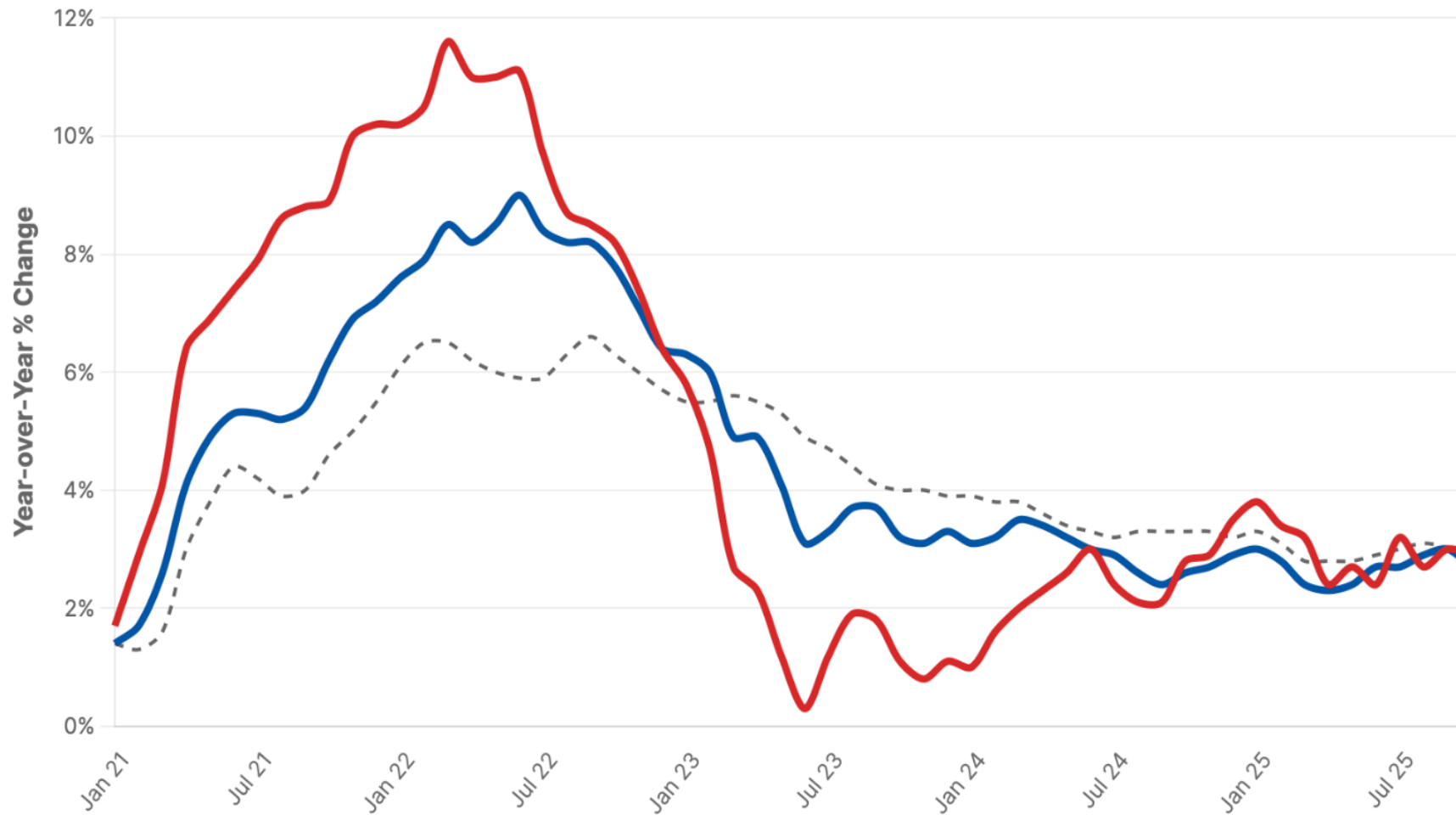
Tracks the cost of a basket of goods and services bought by firms.

Primary use: leading indicator for future consumer prices.

PPI often leads CPI: producer costs get passed on to consumers.

# PPI AS A LEADING INDICATOR

US DATA (2021-2025)



## WHY IT LEADS

When producer costs (PPI) rise, firms eventually pass these costs to consumers.

Notice how the PPI (Red) peaks and turns earlier and more sharply than the CPI (Blue).

- PPI (Producer Price)
- CPI (Consumer Price)
- Core CPI (Ex. Food/Energy)

# How the CPI is Constructed: Five Steps

1

## Fix the Basket

Determine which goods and services are most important to the average consumer.

2

## Find Prices

Collect prices for each item in the basket across locations and outlets.

3

## Compute Basket Cost

Calculate the total cost of the basket for each time period.

4

## Choose Base Year

Select a reference year as the benchmark (Index = 100).

5

## Calculate CPI

$\text{CPI} = (\text{Cost of basket current} / \text{Cost of basket base}) \times 100$ .

### CPI Formula

$\text{CPI} = (\text{Cost of Basket Current} / \text{Cost of Basket Base}) \times 100$

### Inflation Rate

$\text{Inflation} = ((\text{CPI}_2 - \text{CPI}_1) / \text{CPI}_1) \times 100$

# Example: CPI for a Two-Good Economy

 The Basket: 10 Baklavas + 5 Umm Ali

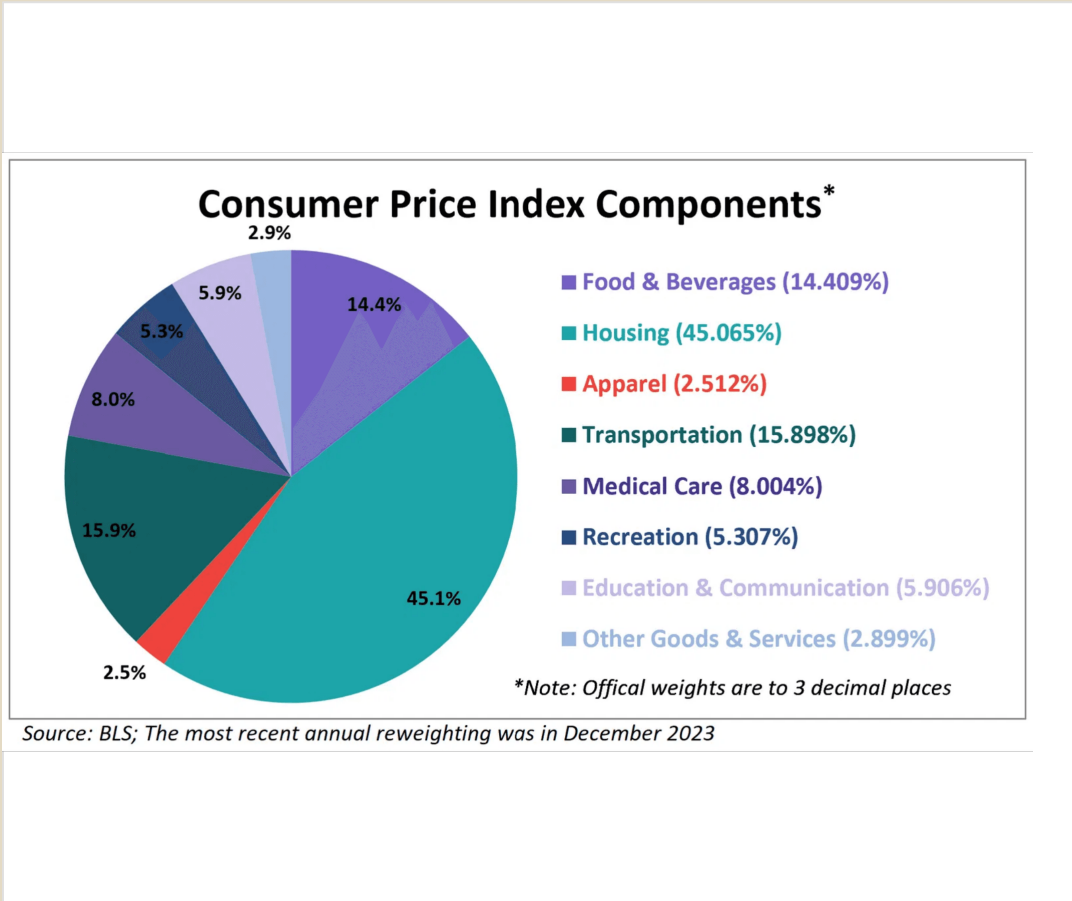
| Year        | Price (Baklava) | Price (Umm Ali) | Cost of Basket Calculation                        | CPI (Base 2024) |
|-------------|-----------------|-----------------|---------------------------------------------------|-----------------|
| 2024 (Base) | QAR 12          | QAR 18          | $(12 \times 10) + (18 \times 5) =$ <b>QAR 210</b> | <b>100.0</b>    |
| 2025        | QAR 14          | QAR 20          | $(14 \times 10) + (20 \times 5) =$ <b>QAR 240</b> | <b>114.3</b>    |
| 2026        | QAR 16          | QAR 22          | $(16 \times 10) + (22 \times 5) =$ <b>QAR 270</b> | <b>128.6</b>    |

**Inflation Rate (2024-2025)**

$((114.3 - 100.0) / 100.0) \times 100$  = **14.3%**



# What's in a Typical CPI Basket?



| Category                  | Weight (US 2023) |
|---------------------------|------------------|
| Housing                   | 34.9%            |
| Transportation            | 16.7%            |
| Food & Beverages          | 14.4%            |
| Medical Care              | 8.1%             |
| Education & Communication | 6.4%             |
| Recreation                | 5.4%             |
| Apparel                   | 2.5%             |
| Other                     | 11.6%            |

Key Insight

Housing is the largest component (approx. 1/3), which is why changes in rent significantly impact overall inflation.

# CPI Basket Weights: Qatar vs US

| Category             | Qatar | United States | Key Difference                             |
|----------------------|-------|---------------|--------------------------------------------|
| Housing              | 21.4% | 34.9%         | Significantly lower housing costs in Qatar |
| Transportation       | 14.8% | 16.7%         | Subsidized fuel in Qatar lowers weight     |
| Food & Beverages     | 13.2% | 14.4%         | Similar importance in both baskets         |
| Education            | 10.5% | 6.4%          | Higher relative spending in Qatar          |
| Recreation & Culture | 8.9%  | 5.4%          | More leisure spending in Qatar basket      |
| Communication        | 5.2%  | -             | Separate category in Qatar                 |
| Other                | 26.0% | 22.2%         | Different categorization methods           |

Think About It



Why might housing have a lower weight in Qatar's CPI basket compared to the US?

Active Learning 02

# Calculating CPI and Inflation

# Active Learning 2

## CPI Calculation Exercise

---

### Scenario

You are an economist at Qatar's Planning and Statistics Authority. Calculate the CPI for a simplified basket:

**Basket: 5 kg Lamb + 10 gal Gasoline + 2 Meals**

| Year | Lamb (per kg) | Gas (per gal) | Meal (each) |
|------|---------------|---------------|-------------|
| 2024 | QAR 40        | QAR 2.00      | QAR 150     |
| 2025 | QAR 44        | QAR 2.20      | QAR 165     |
| 2026 | QAR 50        | QAR 2.50      | QAR 180     |

### Questions

- 1 Calculate the cost of the basket for each year.
- 2 Using 2024 as the base year, calculate the CPI for all three years.
- 3 What was the inflation rate from 2024 to 2025?
- 4 What was the inflation rate from 2025 to 2026?

# Active Learning 2

## Solution

| Year        | Cost Calculation (Lamb + Gas + Meals)              | Total Cost | CPI Calculation          | CPI   |
|-------------|----------------------------------------------------|------------|--------------------------|-------|
| 2024 (Base) | $(5 \times 40) + (10 \times 2.0) + (2 \times 150)$ | QAR 520    | $(520 / 520) \times 100$ | 100.0 |
| 2025        | $(5 \times 44) + (10 \times 2.2) + (2 \times 165)$ | QAR 572    | $(572 / 520) \times 100$ | 110.0 |
| 2026        | $(5 \times 50) + (10 \times 2.5) + (2 \times 180)$ | QAR 635    | $(635 / 520) \times 100$ | 122.1 |

### Inflation Rate (2024-2025)

$$((110.0 - 100) / 100) \times 100$$

**10.0%**

### Inflation Rate (2025-2026)

$$((122.1 - 110) / 110) \times 100$$

**11.0%**

Section 04

# Problems with the CPI

# Three Problems in Measuring the CPI

## **1 Substitution Bias**

The fixed basket ignores that consumers switch toward goods that become relatively cheaper. The CPI overstates the cost of living.

## **2 Introduction of New Goods**

New products raise variety and value, but enter the basket late. The CPI misses part of that gain.

## **3 Unmeasured Quality Change**

If quality rises and the price rises, some of the price increase is not pure inflation. Hard to adjust fully.

# Problem 1: Substitution Bias

When prices rise, consumers substitute toward cheaper alternatives. However, the CPI uses a **fixed basket** and cannot capture this behavior immediately.

Mathematical Impact

↓

Actual spending increase is

lower

(due to substitution).

↑

CPI-measured increase is

higher

(assumes fixed basket).

The Result

The CPI overestimates the increase in the cost of living.



| Scenario             | Consumer Behavior           | CPI Assumption           |
|----------------------|-----------------------------|--------------------------|
| Beef price rises 20% | Buys more chicken (cheaper) | Buys same amount of beef |
| iPhone price rises   | Switches to Android         | Keeps buying iPhone      |
| Gas prices spike     | Carpools or uses transit    | Drives same amount       |



## Problem 2: Introduction of New Goods

- + New goods introduce more variety and choice for consumers.
- \$ More variety means each dollar becomes more valuable (purchasing power rises).

*"Because the CPI uses a **fixed basket**, it does not reflect this increase in the value of the dollar."*

The Result

**The CPI overestimates the cost of living by failing to account for the benefits of new products.**



### Example: The Smartphone

|                                                                  |                  |
|------------------------------------------------------------------|------------------|
| Past (Separate Devices)<br>(Phone + Camera + GPS + Music Player) | \$\$\$ High Cost |
| Present (Smartphone)<br>(All-in-one device)                      | \$ Lower Cost    |

# Problem 3: Unmeasured Quality Change

If the **quality** of a good rises from one year to the next, the **value of a dollar** rises, even if the price of the good stays the same.

## The Economic Logic

- ↑ Quality Improves
- = Consumer gets more value for same price
- ↓ True Cost of Living decreases

## The Result

**Because quality is hard to measure, the CPI often misses these improvements and overestimates inflation.**



## Automobiles

Cars today have airbags, better mileage, and last longer than cars from 20 years ago.

## Medical Care

New treatments and technology improve health outcomes, which is difficult to capture in price data.

## Electronics

Computers and phones become exponentially faster and more capable for similar prices.

Section 05

# Alternatives to the CPI

# Three Alternatives to the CPI

## **1 GDP Deflator**

Broad price index for all domestically produced final goods and services, not a fixed consumer basket.

## **2 PCE Price Index**

- Personal Consumption Expenditures index
- Fed's preferred inflation gauge; allows basket to update.

## **3 Chained / other indexes**

- Indexes that reduce substitution bias by updating weights more often than a classic fixed CPI basket.

# CPI vs GDP Deflator

## CPI



### What it tracks

Prices of goods and services  
*bought by consumers*



### Imports

**INCLUDED** if consumers buy them (e.g.,  
Toyota cars from Japan).



### The Basket

**FIXED** basket of goods.

## GDP Deflator



### What it tracks

Prices of goods and services  
*produced domestically*



### Imports

**EXCLUDED** because they are not part of  
domestic GDP.



### The Basket

**CHANGING** basket (current production).

### Key Difference

Capital goods (like Boeing aircraft) are included in the GDP Deflator (if domestic) but excluded from the CPI.

Active Learning 03

# **CPI vs GDP Deflator Analysis**

# Active Learning 03

## Test Your Understanding

---

In each scenario, determine the effects on the **CPI** and the **GDP Deflator**.



**Starbucks** raises muffin prices.



**Caterpillar** raises tractor prices.



**Armani** raises U.S. jeans prices.

### Ask Yourself:

- ? Is it bought by consumers?
- ? Is it produced domestically?

## 1. Starbucks raises muffin prices

**BOTH RISE**

Muffins are bought by consumers (**CPI**) and produced domestically (**GDP Deflator**).

## 2. Caterpillar raises tractor prices (IL Factory)

**GDP DEFLATOR RISES**

Tractors are capital goods, not consumer goods (No CPI impact), but are produced domestically (**GDP Deflator**).

## 3. Armani raises Italian jeans prices

**CPI RISES**

Jeans are consumer goods (**CPI**), but are imported from Italy, so not part of US production (No GDP Deflator impact).

### Key Insight

The **source** of the good (Domestic vs. Import) and the **buyer** (Consumer vs. Firm) determine which index changes.



Section 06

# Correcting Economic Variables for Inflation

# Comparing Dollar Values Across Time

## The Formula

$$\text{Value in Year 2 Dollars} = \text{Value in Year 1 Dollars} \times (\text{CPI}_2 / \text{CPI}_1)$$

$\text{CPI}_2$  Price level in the target year (current)

$\text{CPI}_1$  Price level in the original year (past)

The inflation correction factor

## Example: Babe Ruth's Salary

In 1931, Babe Ruth earned \$80,000.

CPI in 1931 = 15.2

CPI in 2023 = 307.0

What is his salary in 2023 dollars?

$$\$80,000 \times (307.0 / 15.2)$$

$$\$80,000 \times 20.2$$

$$= \$1,616,000$$

This formula converts "Nominal" figures into "Real" purchasing power

Active Learning 04

# Analyzing Real Costs

A Business Case Study

# Active Learning 04

## Real Tuition Analysis

---

### Tuition & Fees (Nominal Dollars)

| Institution Type            | 1990 Cost | 2018 Cost |
|-----------------------------|-----------|-----------|
| Private Non-Profit (4-Year) | \$9,340   | \$35,830  |
| Public (4-Year)             | \$1,908   | \$10,230  |
| Public (2-Year)             | \$906     | \$3,660   |

CPI 1990

**130.7**

CPI 2018

**251.1**

### Your Task

Express the 1990 tuition figures in **2018 dollars**.

Compute the **percentage increase** in real terms for all three types.

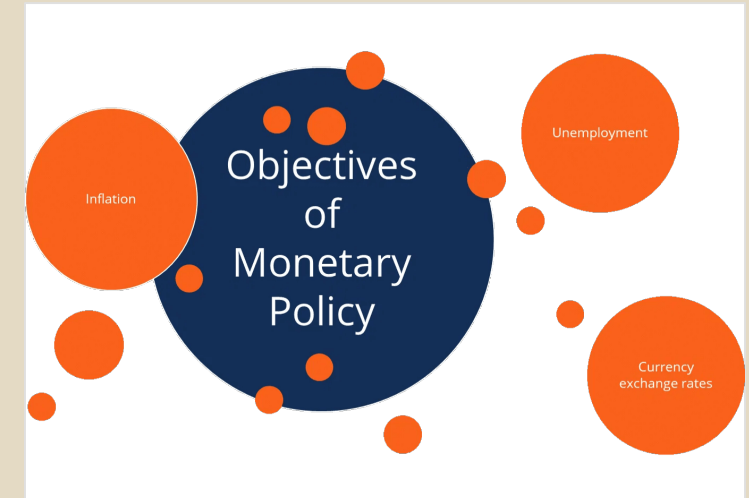
Which type experienced the **largest real increase**?

# Active Learning 4

## Solution

| Institution Type          | 1990<br>(Nominal) | 1990<br>(in<br>2018 \$) | 2018<br>(Nominal) | Real %<br>Change |
|---------------------------|-------------------|-------------------------|-------------------|------------------|
| Private Non-Profit 4-Year | \$9,340           | \$17,944                | \$35,830          | 99.7%            |
| <b>Public 4-Year</b>      | \$1,908           | \$3,665                 | \$10,230          | <b>179.1%</b>    |
| Public 2-Year             | \$906             | \$1,741                 | \$3,660           | 110.3%           |

*\*1990 values adjusted using CPI ratio (251.1 / 130.7 ≈ 1.92)*



### Business Insight

While private colleges are more expensive in absolute terms, **Public 4-Year Universities** have seen the largest *percentage increase* in real cost (nearly tripling).

Section 07

# Important Considerations about Measuring Prices

# Indexation & Regional Differences

## Indexation

The automatic correction by law or contract of a dollar amount for the effects of inflation.

- ✓ **COLA** (Cost of Living Adjustments) in wage contracts
- ✓ **Social Security** benefits
- ✓ **Tax Brackets** (adjusted to prevent "bracket creep")

## Regional Differences

The purchasing power of a dollar is not uniform across geography.

\$100,000 in Mississippi buys significantly more goods and housing than \$100,000 in New York City.

### Business Implication

*"Why could this be important for a company thinking of building a new manufacturing facility?"*  
*Companies like **Amazon** must adjust compensation strategies based on local costs to ensure real wages are competitive.*



# Real vs Nominal Interest Rates

The Fisher Equation (Approx)

$$\text{Real Rate} \approx \text{Nominal Rate} - \text{Inflation}$$



## Nominal Interest Rate

The rate reported by the bank. It tells you how fast the number of dollars in your account rises.



## Real Interest Rate

The rate corrected for inflation. It tells you how fast the *purchasing power* of your account rises.

## The Fisher Effect

In the long run, an increase in the money supply growth rate raises the inflation rate and the nominal interest rate by the same amount.

Nominal

8%

Inflation

- 5%



**Investment Implication:** If inflation is higher than the nominal rate, your real return is negative—you are losing purchasing power.



# Main Takeaways

## **1 Measurement & Alternatives**

CPI is the standard cost-of-living index; Core CPI, PPI, and the GDP deflator answer related but different questions.

## **2 How CPI Is Built**

Fixed basket, prices, cost of basket, index, inflation rate — each step can introduce bias.

## **3 Problems with CPI**

Substitution, new goods, and quality change mean the CPI can overstate inflation.

## **4 Using Price Indexes**

Compare dollars across time; separate real from nominal; watch regional differences and indexation.